Lab 2

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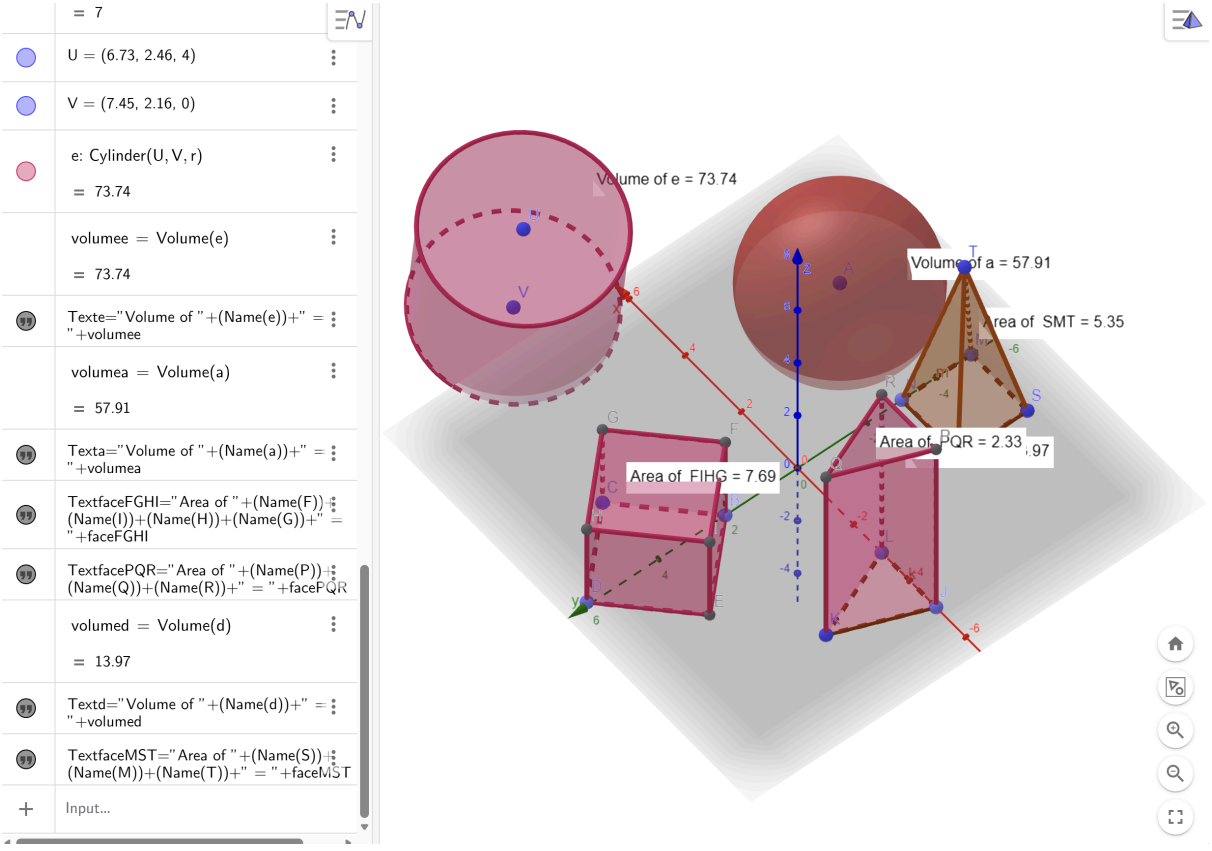
Type	Laboratory
Reviewed	☐

Name	Group	Major
Tănase Vlad Mihai	1	Artificial Intelligence

Task 1

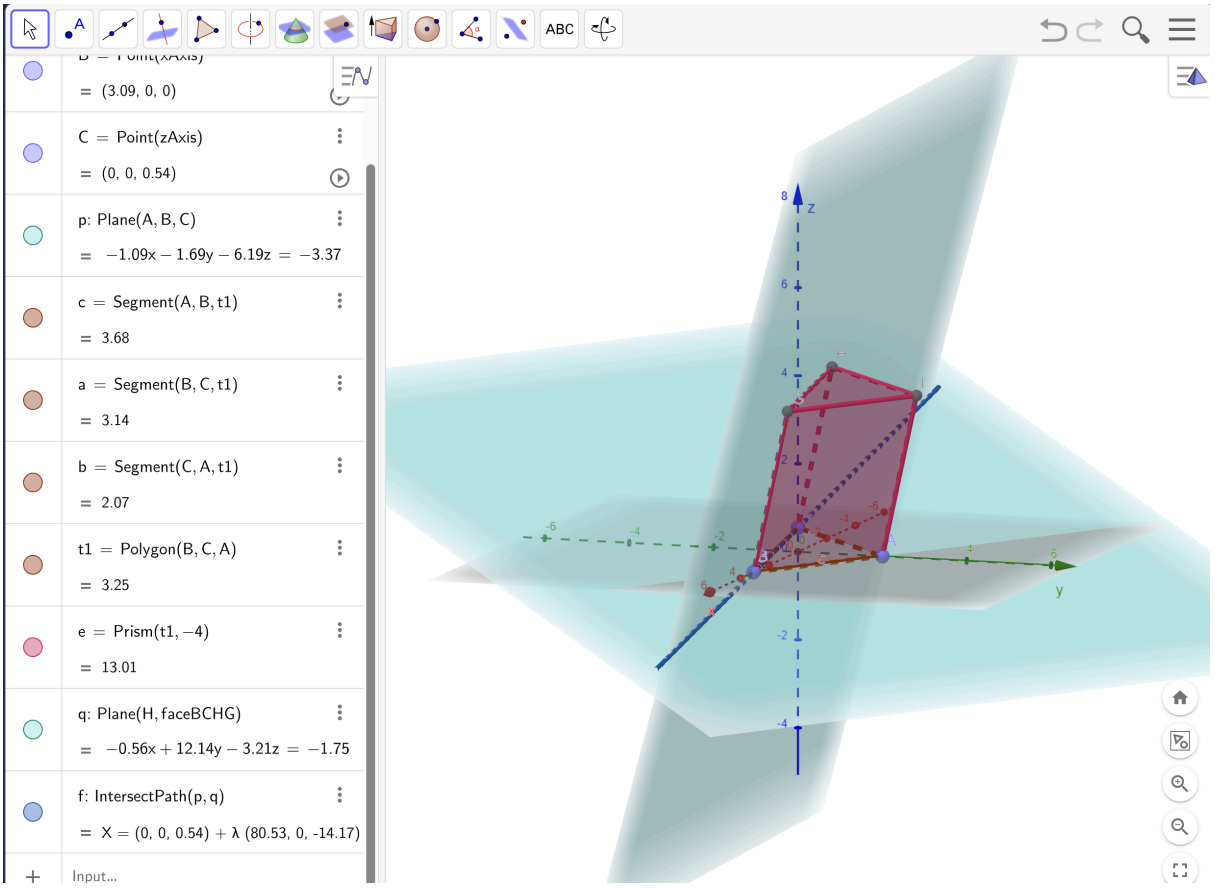
- Create: cube, triangular prism, quadrilateral pyramid, sphere (center + radius),
  cylinder.
- Use commands for Volume/Area where relevant (e.g., Volume[solid]).



https://www.geogebra.org/classic/hyknwvnn

Task 2

- Define a plane through 3 points; create another plane parallel to a face of the
  prism; obtain sections using Intersect.
- Justify the shape of the section (e.g., triangle, rectangle, etc.).

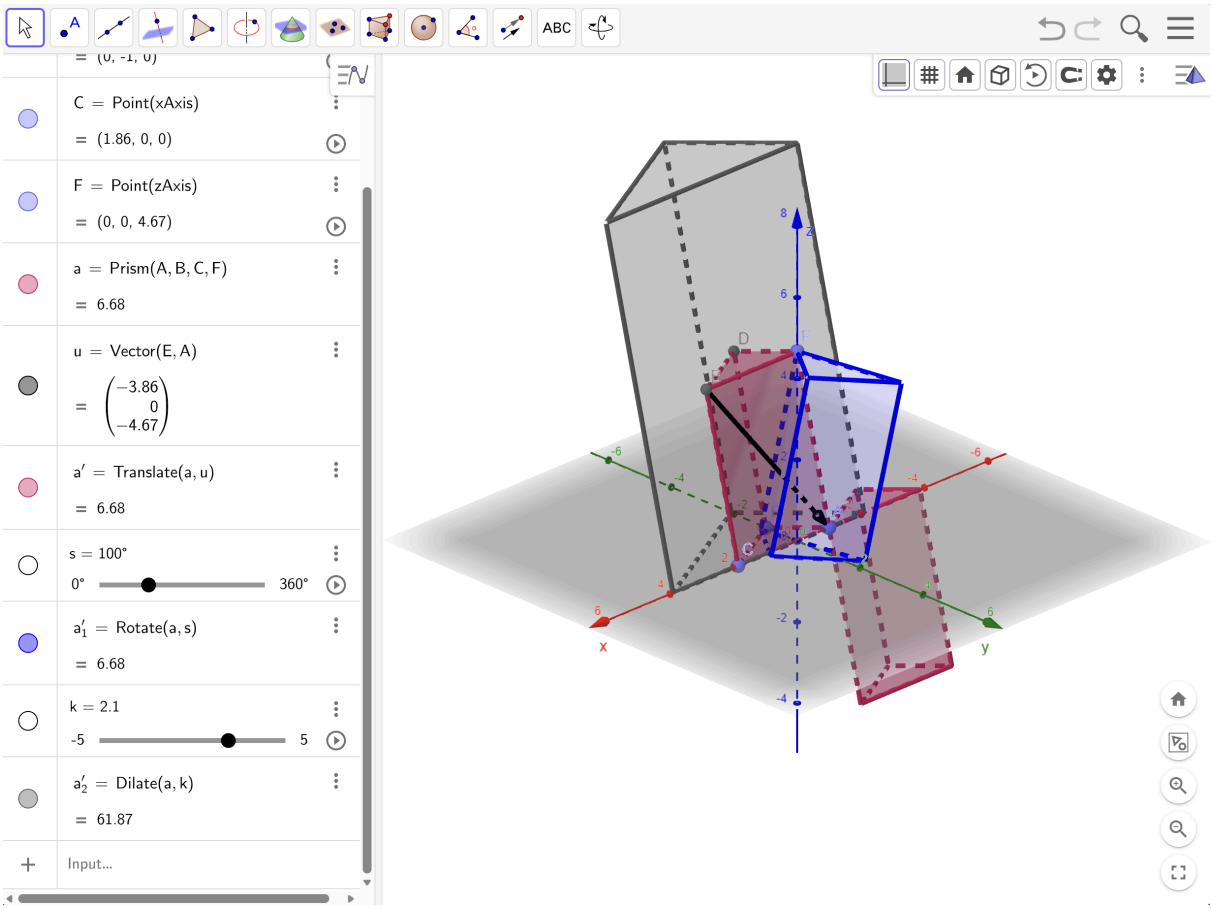


The section is the face BCHG of the prism which is a rectangle.

https://www.geogebra.org/classic/qqnsvq4d

Task 3

- Translate a solid by vector; Rotate (specify angle and axis/center); Dilate with
  scale factor k.
- Use slider(s) for parameters (vector, angle, k) and illustrate the effects.

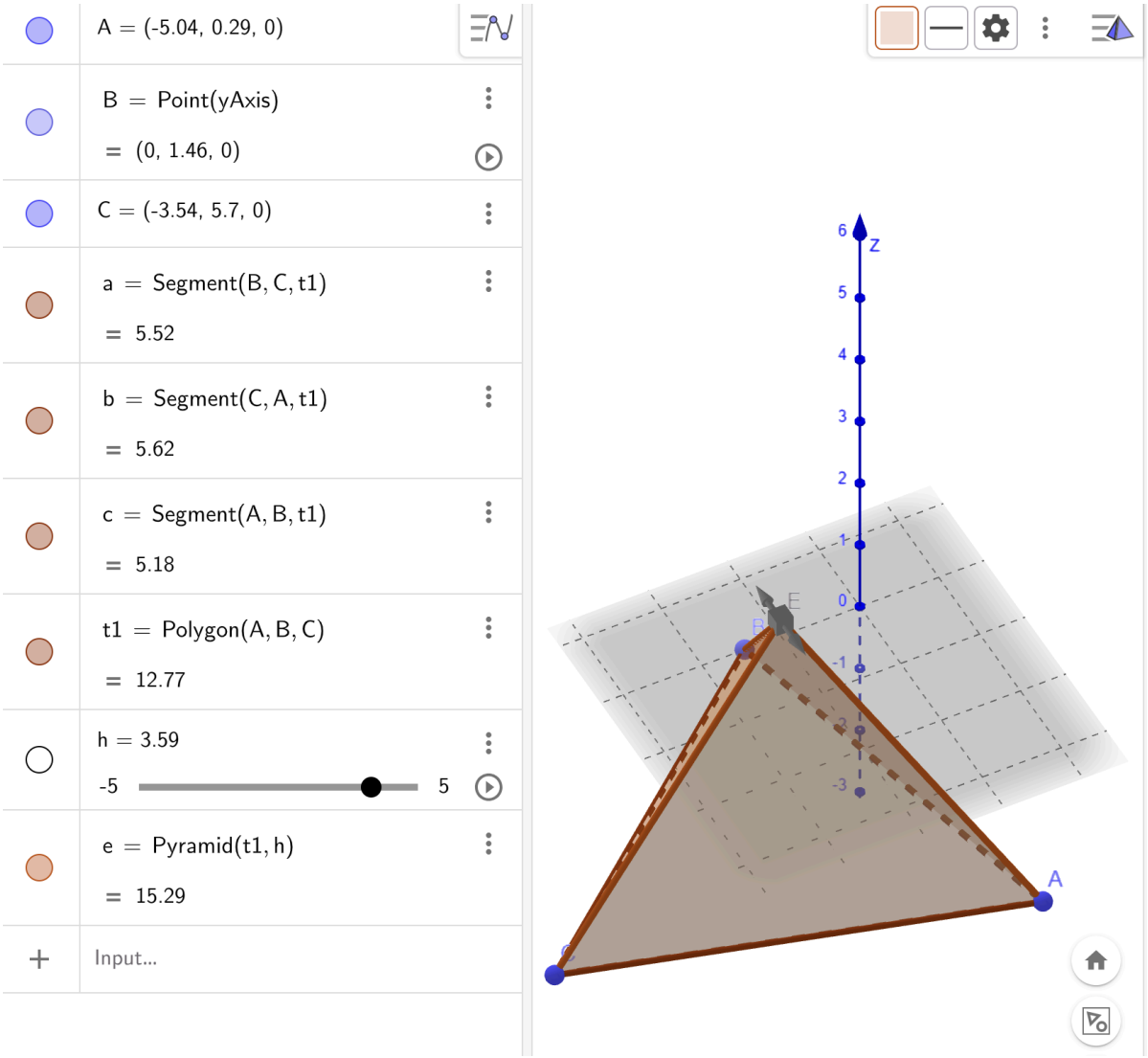


We can see the rotated blue prism of a’1 at angle s=100 degrees and the dilated gray prism a’2 of scale factor k=2.1, all adjusted by sliders

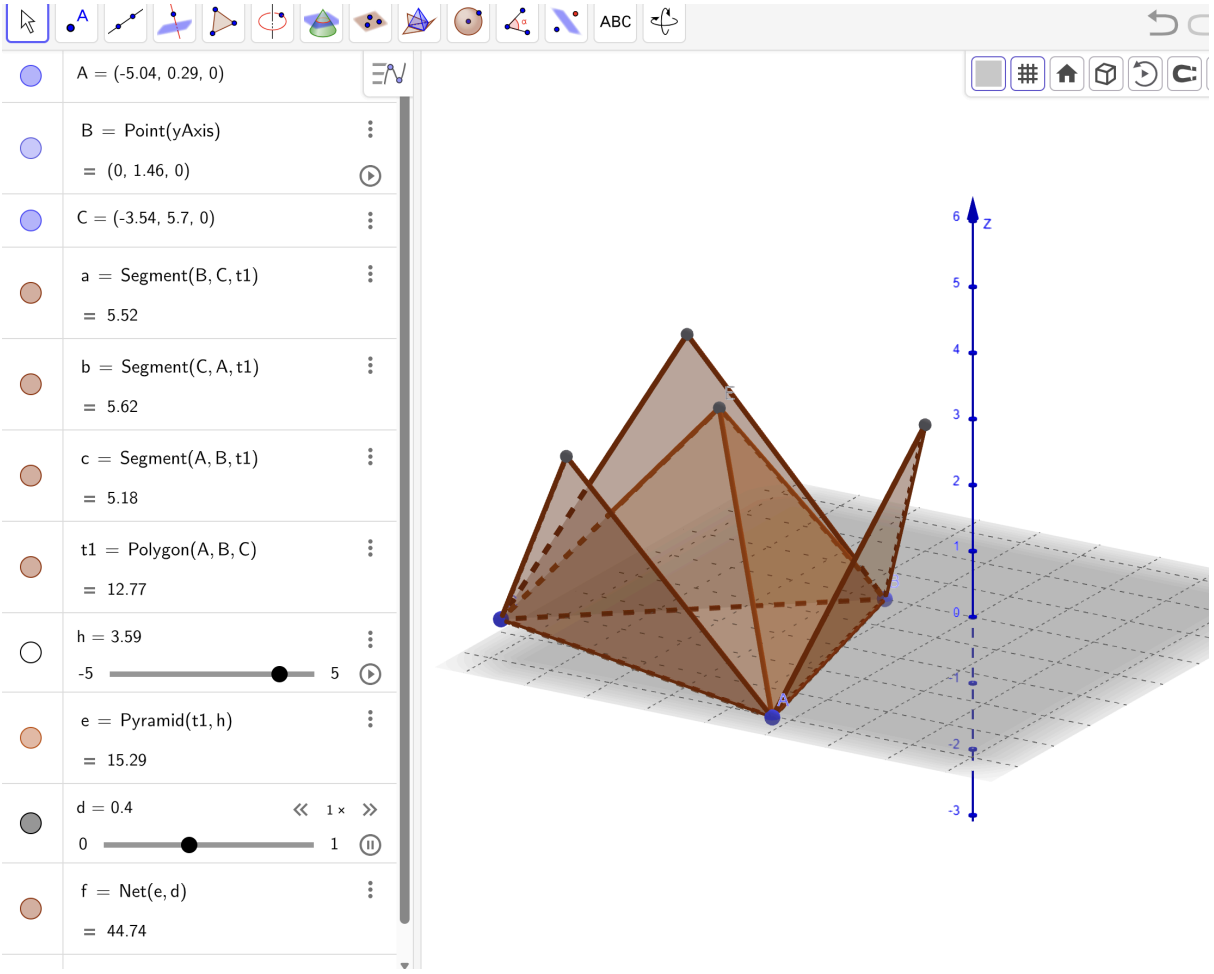
https://www.geogebra.org/classic/u7yejfrx

Task 4

- Adjust the opacity of solids; show/hide axes, xOy-plane, grid.
- Create the net of a prism/pyramid and animate the unfolding.



Created a prism, adjusted its opacity to 20, hid z-y axis only displaying z axis, added grid



Created a net for the prism and then the unfolding is animated by playing the slider for net value d.

https://www.geogebra.org/classic/tpsmdzew